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Procedure Quick-Reference

ORBIT NORTH · Week 3 operator pocket guide

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Procedure Quick-Reference

Twenty-seven procedures students will most likely use during Week 3, in one-card format. The full procedure files are the authoritative source — these cards are pointers, not substitutes.

Card format: **ID · Title · Owner positions · Trigger · Top 3 steps · Exit criterion · File**.

Source of truth for procedures: `~/SpaceMissionSimulation/configs/eosat1/procedures/procedure_index.yaml` plus the per-procedure files in subfolders.

LEOP procedures

Each card follows this template; LEOP-001 is shown in full.

> **LEOP-001 · First Acquisition of Signal & OBC Boot** *Owner positions:* Flight Director · TT&C · FDIR/Systems *Trigger:* First pass over Iqaluit after separation timer expiry. *Top 3 steps:* (1) configure ground station, monitor RF acquisition; (2) verify bootloader beacon (SID 11); (3) issue OBC_BOOT_APP command (S8 func_id 55). *Exit:* App SW running, full HK reporting. *File:* `procedures/leop/first_acquisition.md`

The other LEOP cards, one row each. The trigger is the card's entry condition. Top steps and the exit criterion are on the full card in textbook Part 2, §14.1; the file is the authority. LEOP-003 and LEOP-005 are for advanced students who reach them in the simulation.

ID · Title	Owner positions	Trigger	File
LEOP-002 · Initial Health Check	Flight Director · Power & Thermal · FDIR / Systems	Immediately after LEOP-001; SIDs 1, 3, 4, 6 and 11 streaming	`procedures/leop/initial_health_check.md`
LEOP-007 · Sequential Power-On	Flight Director · Power & Thermal · Flight Dynamics · TT&C · FDIR / Systems	LEOP-004 done (wings deployed); `obdh.sw_image` = 1; SoC > 60 %. Powers only the AOCS suite and commands DETUMBLE	`procedures/leop/sequential_power_on.md`
LEOP-005 · Sun Acquisition	Flight Director · Flight Dynamics	After LEOP-007; AOCS in DETUMBLE (mode 2); SoC > 40 %. Observes the automatic handover to COARSE_SUN	`procedures/leop/sun_acquisition.md`

ID · Title	Owner positions	Trigger	File
LEOP-006 · Time Synchronisation	Flight Director	After LEOP-005; application image running; `ttc.link_status = 2`	`procedures/leop/time_sync.md`
LEOP-003 · Initial Orbit Determination	Flight Director · Flight Dynamics · TT&C	After LEOP-006; COARSE_SUN; GPS fix and SID 12. Last LEOP card: refined TLE to the ground tools, then declares phase 5	`procedures/leop/initial_orbit_determination.md`

Running order under the cold-boot initial state: LEOP-001 → LEOP-002 → LEOP-004 → LEOP-007 → LEOP-005 → LEOP-006 → LEOP-003 (textbook Part 2, §14.1).

Commissioning procedures

The trigger is the card's entry condition. Top steps and the exit criterion are on the full card in textbook Part 2, §14.2. COM-003 (AOCS Sensor Calibration), COM-004 (AOCS Actuator Checkout) and COM-013 are not on this list; see §14.2.

ID · Title	Owner positions	Trigger	File
COM-001 · EPS Checkout	Flight Director · Power & Thermal	Start of commissioning pass 1; COARSE_SUN; SoC > 70 %	`procedures/commissioning/eps_checkout.md`
COM-002 · TCS Verification	Flight Director · Power & Thermal	After COM-001 GO	`procedures/commissioning/tcs_verification.md`
COM-005 · AOCS Mode Transitions	Flight Director · Flight Dynamics	After COM-003 GO (wheels enabled in COM-004, gyros calibrated in COM-003)	`procedures/commissioning/aocs_mode_transitions.md`
COM-006 · TTC Link Verification	Flight Director · TT&C	After COM-005 GO	`procedures/commissioning/ttc_link_verification.md`
COM-007 · OBDH Checkout	Flight Director · FDIR / Systems	Commissioning pass 2, after the pass-startup health check	`procedures/commissioning/obdh_checkout.md`
COM-008 · FDIR Configuration & Protection Functions	Flight Director · FDIR / Systems	After COM-007 GO; FDIR moves from the LEOP set to the nominal set	`procedures/commissioning/fdir_configuration.md`
COM-009 · Payload Power On	Flight Director · Payload Operations ·	Nominal FDIR set armed; SoC > 75 %; generation > 40 W	`procedures/commissioning/payload_power_on.md`

ID · Title	Owner positions	Trigger	File
	Power & Thermal		
COM-010 · FPA Cooler Activation	Flight Director · Payload Operations · Power & Thermal	Immediately after COM-009 GO; payload in STANDBY	`procedures/commissioning/fpa_cooler_activation.md`
COM-011 · Payload Calibration	Flight Director · Payload Operations	FPA in the −16.5…−14.5 °C band; `payload.fpa_ready` = 1	`procedures/commissioning/payload_calibration.md`
COM-012 · First Light	Flight Director · Payload Operations · Flight Dynamics	After COM-011 GO	`procedures/commissioning/first_light.md`

Nominal procedures

The trigger is the card's entry condition. Top steps and the exit criterion are on the full card in textbook Part 2, §14.3.

ID · Title	Owner positions	Trigger	File
NOM-001 · Pass Startup	Flight Director · TT&C	AOS at any of the five network stations	`procedures/nominal/startup.md`
NOM-002 · Imaging Session	Payload Operations · Flight Dynamics	Scheduled imaging window opens; AOCS in or entering FINE_POINT	`procedures/nominal/imaging_session.md`
NOM-003 · Data Downlink	TT&C · Payload Operations	AOS with products queued, usually right after NOM-001	`procedures/nominal/data_downlink.md`
NOM-009 · Routine Health Check	Flight Director	Every pass startup, within NOM-001; or ad hoc by the FD	`procedures/nominal/routine_health_check.md`
NOM-012 · Shift Handover	Flight Director	End of shift	`procedures/nominal/shift_handover.md`

Contingency procedures (the most likely ones for Week 3)

The trigger is the card's entry condition. Top steps and the exit criterion are on the full card in textbook Part 2, §14.4.

ID · Title	Owner positions	Trigger	File
CTG-001 · Under-Voltage Load Shed	Flight Director · Power & Thermal	`eps.uv_flag` (0x010E) = 1, rule ea 1001 fires, or bus below 27.0 V	`procedures/contingency/undervoltage_loadshed.md`
CTG-002 · AOCS Anomaly Recovery	Flight Director · Flight Dynamics	`aocs.att_error` (0x0217) > 5°, a wheel above 60 °C, or a star tracker BLIND	`procedures/contingency/aocs_anomaly.md`
CTG-003 · TTC Link Loss Recovery	Flight Director · TT&C	`ttc.link_status` (0x0501) = 0 for > 30 s after AOS, or the link lost mid-pass	`procedures/contingency/ttc_link_loss.md`
CTG-007 · Reaction Wheel Anomaly	Flight Director · Flight Dynamics	Speed jitter > ±500 rpm over 10 s, speed > 6500 rpm, or bearing above 60 °C	`procedures/contingency/reaction_wheel_anomaly.md`
CTG-012 · Overcurrent Response	Flight Director · Power & Thermal	A trip bit set in `eps.oc_trip_flags` (0x010D)	`procedures/contingency/overcurrent_response.md`
CTG-019 · No Telemetry at Pass Start	Flight Director · TT&C	No beacon and no HK at the expected AOS, or a beacon but no full HK after boot	`procedures/contingency/no_telemetry_at_pass.md`

Emergency procedures (read once, never expected to invoke)

Brief mention only — students should know these exist:

- EMG-001 Emergency Safe Mode (`procedures/emergency/emergency_safe_mode.md`)
- EMG-002 Total Power Failure (`total_power_failure.md`)
- EMG-003 OBC Reboot (`obc_reboot.md`)
- EMG-004 Loss of Communication (`loss_of_communication.md`)
- EMG-005 Loss of Attitude (`loss_of_attitude.md`)
- EMG-006 Thermal Runaway (`thermal_runaway.md`)

If you find yourself reaching for any of these, **call the FD first**. These procedures have safe-mode entry as a step and you do not invoke safe-mode without FD authorisation.

How to read a procedure file

Every procedure file has the same shape; textbook Part 2, §7.4, gives the template. Open `procedures/leop/first_acquisition.md` alongside this:

1. **Header** — `# LEOP-001: First Acquisition of Signal & OBC Boot`, then **Subsystem**, **Phase**, **Revision** and **Approved**. LEOP cards also state an **Entry state** (here: cold boot, OBC in the bootloader).
2. **Purpose** — what the card achieves.
3. **Prerequisites** — a checklist (`- [ ]`) that must all be true before step 1.
4. **Procedure Steps** — `### Step N — name`, built from bold labels rather than a table: **TC** (the command with its arguments inside the backticks, service and subtype beside it), **Action**, **Verify** (mnemonic, hex id, threshold and timeout) and **Note**; some cards add **GO/NO-GO** or **BRANCH**.
5. **Off-Nominal Handling** — what to do when a verification fails, usually naming a CTG or EMG card by ID.
6. **Exit state** and **Post-Conditions** — the exit criterion.

Ownership is not in the file. The positions that run a card are its `required_positions` in `procedure_index.yaml`, which the MCS Procedures tab shows.

How procedures map to MCS commands

Each procedure step that involves commanding lists the PUS service to use (from `mcs/pus_services.yaml`):

Service	Use
S1	Telecommand verification (success / failure of last command)
S3	Housekeeping configuration (which SIDs report when)
S5	Event reports (anomaly events from the spacecraft)
S8	Function management (calling spacecraft functions, e.g. OBC_BOOT_APP)
S9	Time management (set time, time correlation)
S11	Onboard schedule (time-tagged commands)
S15	On-board storage and retrieval (downlink mass memory)
S17	Connection test (uplink-downlink link check)
S19	Event-action service (trigger an OBC action when an event fires)
S20	Parameter management (read / write OBC parameters)

You command via the MCS UI; you do not need to write PUS frames yourself.

When you are stuck

1. Pause. Use a callout: "[Position], standby."
2. Do not command. Re-read the procedure exit criterion.

3. Ask the FD for a 60-second hold.
4. If the FD agrees, log the hold and consult the manual section for the affected subsystem (`configs/eosat1/manual/0*.md`).
5. If you cannot resolve, the FD calls *no-go* and the team rolls back to the last verified state.

References

- Simulator procedure index: `~/SpaceMissionSimulation/configs/eosat1/procedures/procedure_index.yaml`
- ECSS-E-ST-70-32C — Test and operations procedures
- CCSDS Mission Operations Services Concept (CCSDS 520.0-G) — https://public.ccsds.org/Pubs/520x0g4.pdf
- NASA *Space Flight Program and Project Management Requirements* (NPR 7120.5F)